

FINAL EXPLANATION: BIOLOGY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: BIOLOGY GRADE 10: CHEMICALS OF LIFE

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

You have finished all 6 lessons of the Chemicals of Life unit. Now write your own Final Explanation of Wanjiru's blood glucose graph and answer the driving question: What chemicals are in the food we eat, and what happens to them in our bodies? Use your Summary Tables, your first and final flow-diagram models, your Driving Question Board and the evidence from the class investigations. Write in full sentences and use numbers from the data table in your answers. Use these terms correctly: blood glucose concentration (mmol/L), starch, reducing sugar, digestion, absorption, enzyme, amylase, insulin, respiration and glycogen. Here is Wanjiru's data. After fasting, Wanjiru ate a meal of ugali (300 g), sukuma wiki (100 g) and beans (100 g) between 0 and 15 minutes. Her blood glucose was then measured for 3 hours.

Time after meal start (min)	Blood glucose concentration (mmol/L)
0	4.5
15	5.5
30	7.0
60	8.0
90	6.5
120	5.2
150	4.7
180	4.5

The normal fasting range is 4.0 to 5.5 mmol/L.

Part 1: What chemicals are in Wanjiru's lunch?

Wanjiru's lunch was ugali, sukuma wiki and beans. (a)

(a) The meal contains carbohydrates, proteins, lipids (fat from cooking oil), vitamins (A and C in sukuma wiki), water and mineral salts (iron and calcium in sukuma wiki). Carbohydrates, proteins, lipids and vitamins are organic because they contain carbon

Name the groups of chemicals found in this meal and say which are organic and which are inorganic. State the rule you use to decide. (b) Give the approximate amount of each nutrient in the whole meal. (c) Explain which group of chemicals you think is most likely to cause the rise in her blood glucose concentration, and why this is only a prediction at this stage.	bonded to hydrogen. Water and mineral salts are inorganic because they do not. All these chemicals are made from a small set of elements such as C, H, O, N, P, S, Ca, Fe, Mg, Na and K. (b) Ugali has about 70 g carbohydrate, mostly starch from maize. Beans have about 20 g protein and 15 g carbohydrate. Sukuma wiki has about 3 g protein plus fibre, vitamins A and C and mineral salts. The cooking oil gives about 10 g fat. (c) Carbohydrate is the largest amount (about 70 g in ugali), so it is the most likely cause of the rise from 4.5 to 8.0 mmol/L. It is a prediction at this stage because amount alone does not prove which chemical affects blood glucose. We need tests and more evidence.
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Part 2: Food tests as evidence

(a) Describe the iodine test and Benedict's test: what each one detects and what colour results show a positive result. (b) Describe the biuret test and the grease-spot and emulsion tests in the same way. (c) Give the results you recorded for ugali in all of these tests. (d) Ugali gave no change with Benedict's solution before digestion. What does this result, together with what you know about ugali and the glucose in Wanjiru's blood, tell you about what must happen to the starch before glucose can enter the blood?	(a) Iodine detects starch; blue-black means starch is present. Benedict's solution (heated) detects reducing sugars; it goes from blue to green to yellow to orange to brick-red as more reducing sugar is present. Ripe banana gave brick-red/orange and glucose solution gave brick-red. (b) Biuret detects protein; the colour changes from blue to violet/purple (beans, milk and nyama choma were positive). A lipid gives a translucent grease spot on paper, or a cloudy white layer in the emulsion test (groundnuts were positive for both; mandazi gave a translucent spot). (c) Ugali: iodine blue-black (starch present); Benedict's stayed blue (no reducing sugar before digestion); biuret pale blue (negative or weak); lipid tests negative. (d) Ugali contains starch, a large carbohydrate made of many glucose units, but not free reducing sugar. So the starch must be broken down into smaller sugar units before glucose can enter the blood. This links the starch in ugali to the glucose in the blood but shows that a process is needed in between.
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Part 3: Enzymes and digestion

(a) Describe the saliva and starch investigation: what was mixed, the temperature, the tests used and the control. (b) Give the results at 0, 5 and 10 minutes and for the control. (c) Explain these results using the words enzyme, amylase,	(a) 2 cm³ of 1% starch solution was mixed with 1 cm³ of saliva in a test tube kept at about 37 degrees C in a warm water bath. Iodine was used to test samples at 0, 5 and 10 minutes, and Benedict's test was done on the 10 minute sample. The control used boiled saliva or water. (b) Iodine was blue-black at 0 min, brown-black at 5 min and yellow-brown at 10 min, so no starch was left. Benedict's at 10 min was brick-red, so reducing sugar was present. The control stayed blue-black at 10 min. (c) Saliva contains amylase, an enzyme. Enzymes are biological catalysts and are proteins. Amylase digests starch (large and insoluble) to maltose, which is then broken to glucose, both small and soluble. This is digestion. Starch disappeared and reducing sugar appeared only where active amylase was present. (d) Enzymes work best at about 37 degrees C in the body. Boiling denatures amylase, which is why the boiled-saliva control stayed blue-black. At 0 degrees C the enzyme is inactive. (e) Water and mineral salts are inorganic
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digestion and reducing sugar. (d) Explain what temperature does to amylase and why 37 degrees C matters. (e) Say what role water and mineral salts play.	chemicals that support digestion; water provides the medium in which the chemicals dissolve and react.
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Part 4: Explaining the rise in blood glucose

Use the data table. (a) Calculate the rise in blood glucose concentration from 0 to 60 minutes, and the percentage it is above the starting level. (b) Explain, step by step, why the concentration rose after Wanjiru's meal, using the words digestion and absorption. (c) Explain why you think carbohydrates, and not proteins or lipids, are the main cause of the spike. Use evidence from the meal and from the tests.	(a) The rise is $8.0 - 4.5 = 3.5$ mmol/L in 60 minutes. This is $3.5 / 4.5$, about 78% above the starting level. (b) The starch in the ugali (and smaller amounts of other carbohydrates) is digested by enzymes such as amylase into glucose. The glucose is then absorbed into the blood. As more glucose moves into the blood than leaves it, the blood glucose concentration climbs from 4.5 mmol/L, through 5.5 at 15 min and 7.0 at 30 min, to the peak of 8.0 mmol/L at 60 minutes. (c) The meal had about 70 g of carbohydrate in the ugali (plus about 15 g in the beans), compared with about 20 g protein and about 10 g fat. Ugali tested positive for starch, and starch is made of glucose units. Proteins are made of amino acids and lipids of fatty acids and glycerol, so they are not built from glucose. They are used for growth, repair and energy storage, so carbohydrates are the main cause of the spike.
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Part 5: Explaining the fall and the whole story

(a) Calculate the fall in blood glucose concentration from 60 to 180 minutes and compare how fast the rise and the fall happened. (b) State when the level was first back in the normal fasting range. (c) Explain what the body does to bring blood glucose back down, naming the organ and hormone involved and what cells, the liver and muscles do with glucose. (d) Write the equation for respiration in words. (e) Finish by writing a short answer to the driving question: What chemicals are in the food we eat, and what happens to them	(a) The fall is $8.0 - 4.5 = 3.5$ mmol/L over 120 minutes (60 to 180 min). The rise was 3.5 mmol/L in 60 minutes, so the rise was faster than the fall. (b) The level was first back in the normal fasting range of 4.0 to 5.5 mmol/L at 120 minutes, at 5.2 mmol/L, and it stayed in the range at 150 min (4.7) and 180 min (4.5). (c) The pancreas releases the hormone insulin, which lowers blood glucose. Insulin causes cells, the liver and muscles to take up glucose from the blood. Cells use glucose in respiration to release energy, and the liver and muscles store the excess as glycogen. As a result the concentration falls: 6.5 at 90 min, 5.2 at 120 min, 4.7 at 150 min and 4.5 mmol/L at 180 min. (d) Glucose + oxygen gives carbon dioxide + water + energy (ATP released). (e) Wanjiru's lunch contains organic chemicals (carbohydrates, proteins, lipids, vitamins) and inorganic chemicals (water and mineral salts). Starch in ugali is digested by enzymes such as amylase into glucose, which is absorbed into the blood and raises blood glucose to 8.0 mmol/L at 60 min. Insulin from the pancreas then makes cells, the liver and muscles take up glucose, which is used in respiration or stored as glycogen, so the level returns to 4.5 mmol/L by 180 min. Proteins and lipids from the beans, sukuma wiki and cooking oil are also digested and used for growth, repair and energy storage, but carbohydrates are the main cause of the spike.
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in our bodies?	
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Chemicals of life (composition and classification)	Correctly names all the food chemical groups, classifies them as organic or inorganic using the carbon-bonded-to-hydrogen rule, names the building blocks and links them to the meal amounts.	Names most chemical groups and classifies them correctly, with some building blocks or meal amounts missing.	Names few groups, or confuses organic and inorganic, with little reference to the meal.
Use of evidence from investigations	Uses specific test results (colours, foods, times, mmol/L values) accurately from several lessons to support claims, including the ugali Benedict's result.	Uses some specific results accurately but misses some or states them in general terms.	Uses little evidence, or results are vague or incorrect.
Enzymes and digestion	Explains amylase as a protein catalyst digesting starch to maltose then glucose, uses the saliva results and the boiled-saliva control, and explains the effect of temperature (37 degrees C, denaturation, inactive at 0 degrees C).	Explains the role of amylase and the main results, with incomplete reasoning about temperature or the control.	Mentions enzymes but does not explain how they digest starch or how temperature affects them.
Explaining the phenomenon (rise and fall)	Gives a complete sequence: digestion, absorption, rise to 8.0 mmol/L at 60 min, insulin, uptake, respiration and glycogen storage, fall to 4.5 mmol/L at 180 min, with correct calculations.	Explains both the rise and the fall with minor gaps in the sequence or calculations.	Explains only the rise or only the fall, or has major gaps or calculation errors.
Scientific reasoning	Links claims to evidence and gives reasons why carbohydrates, not proteins or lipids, are the main cause of the spike; writes clearly using the correct terms.	Gives some reasoning linking evidence to claims, with some unsupported statements.	Mostly lists facts without reasoning or evidence.

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